TBDA G2 – Model parameters, tuning grid and results (dataset v1.2, grid v1.3)

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Summary

The extended v1.3 grid (591 parameter combinations per horizon, at least 100 per model) was run on the validation week (22–28 Sep 2025). With the selection rule D10 agreed beforehand, the **blend of GBM, ChronosGBM and Chronos** is the final model at all three horizons. It beats the best baseline clearly at every horizon, including day-ahead. Every configuration and the model choice are frozen in `models/*_frozen_config_v1.3.json` and `models/model_choice_v1.3.json`.

At day-ahead this is a clear change from v1.2: persistence (B3, 0.3708) was then only 0.0047 behind the best model (Chronos, 0.3661), inside the tie margin. With v1.3 the final blend (0.3513) beats it by 0.0195, about four times the tie margin.

Horizon	Final model	Weights GBM / ChronosGBM / Chronos	Validation score (scaled MAE)	Best baseline (gap)
+1h	ML_Blend	0.30 / 0.20 / 0.50	0.2712	B3 0.3586 (−0.087)
+6h	ML_Blend	0.10 / 0.35 / 0.55	0.3488	B2 0.5211 (−0.172)
day-ahead	ML_Blend	0.15 / 0.10 / 0.75	0.3513	B3 0.3708 (−0.019)

The best single models are ChronosGBM at +1h (0.2665) and +6h (0.3461), and Chronos-Bolt Base at day-ahead (0.3477). The blend is within the tie margin of 0.005 of each of them and has the lower scaled RMSE, so D10 chooses it (see Interpretation).

Status: validation results only. The single test run (29 Sep – 5 Oct 2025, point 8) has not been done. The model choice is fixed before any test forecast exists. All figures come from the v1.3 runs of 29 Oct 2026 and were recomputed independently from the prediction files.

Protocol

Every model is fitted on training data only, compared on the validation week with one fixed metric, and frozen before the test week is used. Nothing in the protocol changed from v1.2 except the grids and the selection rule D10.

Item	Value
Target	`energy_Wh` per hour, 16 Storey 1 circuits (all with 2025 data)
Dataset	v1.2 (unchanged)
Train	19 Aug – 21 Sep 2025 (UTC)
Validation	22 – 28 Sep 2025, 168 h
Test	29 Sep – 5 Oct 2025, 168 h, used once (point 8)
Horizons	+1h (lead 1), +6h (lead 6), day-ahead (leads 1–24, issued 00:00 Madrid)
Selection metric	mean over the 15 training circuits of validation MAE ÷ circuit training mean ("scaled MAE"), each model on its own forecasts

Item	Value
Selection rule	D10 (next section), agreed 29 Oct 2026 before the first v1.3 run
Training circuits	15; SE Plugs & Sockets is forecast and scored as cold-start
Seed	0

Selection rule D10

D10 is used twice: to choose the configuration inside every grid, and to choose the final model per horizon among the four ML models.

1. Score = scaled MAE over the 15 training circuits on validation (as in v1.2).
2. Every candidate within **0.005** of the best score is tied with it (D10b).
3. A tied candidate whose forecasts are identical to a simpler one's is replaced by the simpler one.
4. Among the tied candidates the lowest **mean scaled RMSE** wins. Equal RMSE: the simpler model.

D10a: baselines (B1–B3) cannot be chosen; only Chronos, GBM, ChronosGBM and Blend are candidates.
D10c: the baselines are tracked in every tuning log (their scores on the same rows, the gap to the best baseline, and whether a configuration beats it by more than 0.005), so every parameter combination can be compared with them.

Parameter grid (v1.3)

Every parameter that is tuned, with all values tried. Every combination was run on all three horizons.

Model	Parameter	Values	Count
GBM	learning_rate	0.03, 0.05, 0.1, 0.2	4
GBM	max_leaf_nodes	7, 15, 31, 63	4
GBM	min_samples_leaf	10, 20, 50, 100	4
GBM	l2_regularization	0, 1	2
GBM	combinations	4 × 4 × 4 × 2	128
ChronosGBM	same four parameters	same values as GBM; Chronos inputs from the frozen v1.3 Chronos	128
Chronos-Bolt	model size	Tiny, Mini, Small, Base (zero-shot)	4
Chronos-Bolt	context length (h)	24, 48, 72, 96, 120, 168, 240, 336, 420, 504, 588, 672, 816	13
Chronos-Bolt	point forecast	median, mean of the 9 quantiles 0.1–0.9	2
Chronos-Bolt	combinations	4 × 13 × 2	104
Blend	weights GBM / ChronosGBM / Chronos	every mix on a 0.05 grid that sums to 1 (includes each model alone and every two-model mix)	231
All four	combinations per horizon	128 + 128 + 104 + 231	591

Fixed settings (not tuned, same for all runs):

Model	Parameter	Value
GBM, ChronosGBM	mode	pooled (one model, circuit as category); won every horizon in v1.1 and v1.2
GBM, ChronosGBM	loss / max_iter / early stopping	absolute_error / 300 / off
GBM	inputs	15 features: last_val, mean_24, std_24, mean_168, lag_T24, lag_T168, wk_median, last_age_h, n_24, n_168, n_weeks, hour, weekday, is_weekend, lead
ChronosGBM	inputs	GBM features + chr_q10, chr_q50, chr_q90, chr_width (÷ circuit scale), chr_ctx_h
Chronos-Bolt	prediction length / clipping	24 h / forecasts clipped at 0
Baselines	B1 / B2 / B3	lag 168 h / median per weekday × hour / last valid hour, max. 24 h old

Data parameters (gap rule 15 min, consistency ratio 2 and tolerance 1 Wh, training rule 168 h and 50 % flagged) are unchanged from v1.2 and listed in `README_parameters.md`.

Tuning results

For each model: the range of validation scores over the grid, the five best configurations per horizon and the one D10 chose (highlighted). The full logs with every combination, per-circuit scores and the baseline columns are `results/*_tuning_log_v1.3.csv`.

GBM

Horizon	Range (best – worst)	Tied (D10)	Clearly beat best baseline	Best baseline
+1h	0.2817 – 0.4081	4	124 / 128	B3 persistence 0.3586
+6h	0.3975 – 0.4780	1	128 / 128	B2 weekday-hour median 0.5211
day-ahead	0.3970 – 0.4720	4	0 / 128	B3 persistence 0.3708

Horizon	Rank	learning rate	leaves	min leaf	L2	Scaled MAE	Scaled RMSE	D10
+1h	1	0.1	31	50	0	0.2817	0.6396	tied
+1h	2	0.1	15	50	1	0.2824	0.6341	chosen
+1h	3	0.03	63	50	1	0.2855	0.6411	tied
+1h	4	0.1	15	10	0	0.2866	0.6370	tied
+1h	5	0.05	63	50	1	0.2877	0.6466	
+6h	1	0.1	7	10	1	0.3975	0.7509	chosen
+6h	2	0.03	7	50	0	0.4081	0.7619	
+6h	3	0.05	7	100	0	0.4087	0.7687	
+6h	4	0.03	7	100	0	0.4090	0.7635	

Horizon	Rank	learning rate	leaves	min leaf	L2	Scaled MAE	Scaled RMSE	D10
+6h	5	0.05	7	100	1	0.4095	0.7662	
day-ahead	1	0.05	7	20	0	0.3970	0.7497	chosen
day-ahead	2	0.05	7	20	1	0.3982	0.7509	tied
day-ahead	3	0.05	7	10	1	0.4001	0.7520	tied
day-ahead	4	0.05	7	50	0	0.4019	0.7545	tied
day-ahead	5	0.05	7	50	1	0.4039	0.7565	

At +6h and day-ahead the smallest trees (7 leaves) win; the mean score over all configurations rises with every step in tree size. At +1h tree size matters little. At day-ahead **no GBM configuration beats persistence** (B3 0.3708; best GBM 0.3970).

ChronosGBM

Horizon	Range (best – worst)	Tied (D10)	Clearly beat best baseline	Best baseline
+1h	0.2665 – 0.3494	5	128 / 128	B3 persistence 0.3586
+6h	0.3461 – 0.4922	4	128 / 128	B2 weekday-hour median 0.5211
day-ahead	0.3762 – 0.4715	4	0 / 128	B3 persistence 0.3708

Horizon	Rank	learning rate	leaves	min leaf	L2	Scaled MAE	Scaled RMSE	D10
+1h	1	0.03	63	50	0	0.2665	0.6197	chosen
+1h	2	0.03	63	20	0	0.2672	0.6205	tied
+1h	3	0.03	63	50	1	0.2686	0.6243	tied
+1h	4	0.1	63	50	0	0.2692	0.6238	tied
+1h	5	0.03	31	50	0	0.2712	0.6209	tied
+6h	1	0.05	7	50	1	0.3461	0.7185	chosen
+6h	2	0.05	7	100	1	0.3496	0.7192	tied
+6h	3	0.05	7	50	0	0.3499	0.7195	tied
+6h	4	0.03	15	100	0	0.3505	0.7192	tied
+6h	5	0.05	7	100	0	0.3512	0.7188	
day-ahead	1	0.03	15	100	1	0.3762	0.7278	chosen
day-ahead	2	0.05	7	100	0	0.3783	0.7309	tied
day-ahead	3	0.2	7	50	1	0.3798	0.7392	tied
day-ahead	4	0.05	7	100	1	0.3801	0.7310	tied
day-ahead	5	0.1	7	100	0	0.3817	0.7348	

The Chronos inputs make the GBM much better at +1h and +6h. At +6h and day-ahead large `min_samples_leaf` (50–100) and few leaves help, as for the GBM. At day-ahead no configuration beats persistence (best 0.3762).

Chronos-Bolt

Horizon	Range (best – worst)	Tied (D10)	Clearly beat best baseline	Best baseline
+1h	0.2804 – 0.3137	12	104 / 104	B3 persistence 0.3586
+6h	0.3510 – 0.4111	13	104 / 104	B2 weekday-hour median 0.5211
day-ahead	0.3477 – 0.3964	2	32 / 104	B3 persistence 0.3708

Horizon	Rank	size	context (h)	point	Scaled MAE	Scaled RMSE	D10
+1h	1	base	588	median	0.2804	0.6238	tied
+1h	2	base	72	median	0.2813	0.6272	tied
+1h	3	base	96	median	0.2813	0.6226	tied
+1h	4	base	168	median	0.2820	0.6315	tied
+1h	5	base	672	median	0.2820	0.6177	tied
+1h	7	base	588	mean of quantiles	0.2832	0.6142	chosen
+6h	1	mini	504	median	0.3510	0.7159	tied
+6h	2	mini	336	median	0.3519	0.7194	tied
+6h	3	small	816	median	0.3521	0.7156	tied
+6h	4	mini	588	median	0.3521	0.7172	tied
+6h	5	small	504	median	0.3525	0.7195	tied
+6h	13	tiny	24	median	0.3554	0.7113	chosen
day-ahead	1	base	72	median	0.3477	0.7059	chosen
day-ahead	2	mini	72	median	0.3503	0.7085	tied
day-ahead	3	base	24	median	0.3531	0.7078	
day-ahead	4	base	168	median	0.3536	0.7270	
day-ahead	5	small	72	median	0.3540	0.7114	

The median beats the mean of the quantiles on average at every horizon. Base is the best size at +1h and day-ahead, at +6h all sizes lie within 0.007. At day-ahead a short context of 72 h is best; Chronos then follows level changes faster. At +1h and +6h D10 picked a tied configuration with a lower RMSE but a slightly higher MAE than the best one (+0.0028 and +0.0044) (Base 588 h mean of quantiles at +1h, Tiny 24 h at +6h).

Blend

Horizon	Range (best – worst)	Tied (D10)	Clearly beat best baseline	Best baseline
+1h	0.2664 – 0.2832	116	231 / 231	B3 persistence 0.3586

Horizon	Range (best – worst)	Tied (D10)	Clearly beat best baseline	Best baseline
+6h	0.3439 – 0.3975	48	231 / 231	B2 weekday-hour median 0.5211
day-ahead	0.3474 – 0.3970	27	125 / 231	B3 persistence 0.3708

Horizon	Rank	w GBM	w ChronosGBM	w Chronos	Scaled MAE	Scaled RMSE	D10
+1h	1	0.05	0.90	0.05	0.2664	0.6174	tied
+1h	2	0.10	0.85	0.05	0.2664	0.6166	tied
+1h	3	0.10	0.80	0.10	0.2665	0.6154	tied
+1h	4	0.05	0.85	0.10	0.2665	0.6162	tied
+1h	5	0.00	0.95	0.05	0.2665	0.6184	tied
+1h	113	0.30	0.20	0.50	0.2712	0.6078	chosen
+6h	1	0.05	0.75	0.20	0.3439	0.7115	tied
+6h	2	0.05	0.70	0.25	0.3439	0.7102	tied
+6h	3	0.05	0.80	0.15	0.3440	0.7128	tied
+6h	4	0.05	0.65	0.30	0.3440	0.7092	tied
+6h	5	0.05	0.85	0.10	0.3441	0.7144	tied
+6h	48	0.10	0.35	0.55	0.3488	0.7059	chosen
day-ahead	1	0.00	0.05	0.95	0.3474	0.7050	tied
day-ahead	2	0.00	0.00	1.00	0.3477	0.7059	tied
day-ahead	3	0.00	0.10	0.90	0.3478	0.7043	tied
day-ahead	4	0.05	0.05	0.90	0.3479	0.7039	tied
day-ahead	5	0.05	0.00	0.95	0.3480	0.7047	tied
day-ahead	23	0.15	0.10	0.75	0.3513	0.7027	chosen

Many mixes are tied (116 of 231 at +1h), so the exact weights matter little. The best-scoring mixes are dominated by ChronosGBM at +1h and +6h and by Chronos at day-ahead; D10 chose more balanced mixes with lower RMSE.

Effect of each parameter

Mean validation score over all configurations that share a value (lower is better). This shows the direction of each parameter, not the best configuration.

Model	Parameter	Value	+1h	+6h	day-ahead
GBM	learning rate	0.03	0.3077	0.4318	0.4324
GBM	learning rate	0.05	0.3204	0.4306	0.4330
GBM	learning rate	0.1	0.3190	0.4391	0.4370
GBM	learning rate	0.2	0.3249	0.4487	0.4440
GBM	leaves	7	0.3184	0.4135	0.4125
GBM	leaves	15	0.3261	0.4391	0.4398

Model	Parameter	Value	+1h	+6h	day-ahead
GBM	leaves	31	0.3114	0.4450	0.4442
GBM	leaves	63	0.3160	0.4525	0.4500
GBM	min leaf	10	0.3158	0.4399	0.4299
GBM	min leaf	20	0.3131	0.4407	0.4345
GBM	min leaf	50	0.3139	0.4404	0.4399
GBM	min leaf	100	0.3291	0.4292	0.4421
GBM	L2	0	0.3171	0.4373	0.4363
GBM	L2	1.0	0.3188	0.4378	0.4370
ChronosGBM	learning rate	0.03	0.2839	0.3928	0.4139
ChronosGBM	learning rate	0.05	0.2869	0.3994	0.4144
ChronosGBM	learning rate	0.1	0.2937	0.4117	0.4180
ChronosGBM	learning rate	0.2	0.3050	0.4211	0.4254
ChronosGBM	leaves	7	0.3059	0.3656	0.3966
ChronosGBM	leaves	15	0.2897	0.4123	0.4229
ChronosGBM	leaves	31	0.2859	0.4256	0.4245
ChronosGBM	leaves	63	0.2881	0.4216	0.4277
ChronosGBM	min leaf	10	0.2944	0.4289	0.4330
ChronosGBM	min leaf	20	0.2897	0.4300	0.4187
ChronosGBM	min leaf	50	0.2885	0.3979	0.4287
ChronosGBM	min leaf	100	0.2970	0.3683	0.3912
ChronosGBM	L2	0	0.2921	0.4058	0.4185
ChronosGBM	L2	1.0	0.2927	0.4067	0.4173
Chronos-Bolt	size	tiny	0.3010	0.3697	0.3694
Chronos-Bolt	size	mini	0.2958	0.3648	0.3668
Chronos-Bolt	size	small	0.2932	0.3632	0.3677
Chronos-Bolt	size	base	0.2862	0.3700	0.3739
Chronos-Bolt	point forecast	median	0.2913	0.3589	0.3663
Chronos-Bolt	point forecast	mean of quantiles	0.2968	0.3750	0.3726

Best Chronos score per context length (any size and point forecast):

Context (h)	+1h	+6h	day-ahead
24	0.2931	0.3528	0.3531
48	0.2839	0.3568	0.3574
72	0.2813	0.3582	0.3477
96	0.2813	0.3590	0.3631
120	0.2821	0.3580	0.3614
168	0.2820	0.3570	0.3536
240	0.2847	0.3565	0.3694
336	0.2867	0.3519	0.3676

Context (h)	+1h	+6h	day-ahead
420	0.2859	0.3554	0.3682
504	0.2851	0.3510	0.3618
588	0.2804	0.3521	0.3627
672	0.2820	0.3526	0.3607
816	0.2835	0.3521	0.3626

Best parameter combination per model

The frozen configuration of each model, per horizon. These are the settings that go into the test run.

Model	Horizon	Chosen parameters (D10)	Validation score	Scaled RMSE	Tied
GBM	+1h	lr 0.1, 15 leaves, min leaf 50, L2 1	0.2824	0.6341	4
GBM	+6h	lr 0.1, 7 leaves, min leaf 10, L2 1	0.3975	0.7509	1
GBM	day-ahead	lr 0.05, 7 leaves, min leaf 20, L2 0	0.3970	0.7497	4
Chronos-Bolt	+1h	Base, context 588 h, mean of quantiles	0.2832	0.6142	12
Chronos-Bolt	+6h	Tiny, context 24 h, median	0.3554	0.7113	13
Chronos-Bolt	day-ahead	Base, context 72 h, median	0.3477	0.7059	2
ChronosGBM	+1h	lr 0.03, 63 leaves, min leaf 50, L2 0; Chronos Base 588 h	0.2665	0.6197	5
ChronosGBM	+6h	lr 0.05, 7 leaves, min leaf 50, L2 1; Chronos Tiny 24 h	0.3461	0.7185	4
ChronosGBM	day-ahead	lr 0.03, 15 leaves, min leaf 100, L2 1; Chronos Base 72 h	0.3762	0.7278	4
Blend	+1h	0.30 GBM + 0.20 ChronosGBM + 0.50 Chronos	0.2712	0.6078	116
Blend	+6h	0.10 GBM + 0.35 ChronosGBM + 0.55 Chronos	0.3488	0.7059	48
Blend	day-ahead	0.15 GBM + 0.10 ChronosGBM + 0.75 Chronos	0.3513	0.7027	27

All GBM and ChronosGBM models: pooled mode, absolute-error loss, 300 iterations, no early stopping, seed 0, fitted on training rows whose target hour lies before the first validation issue time.

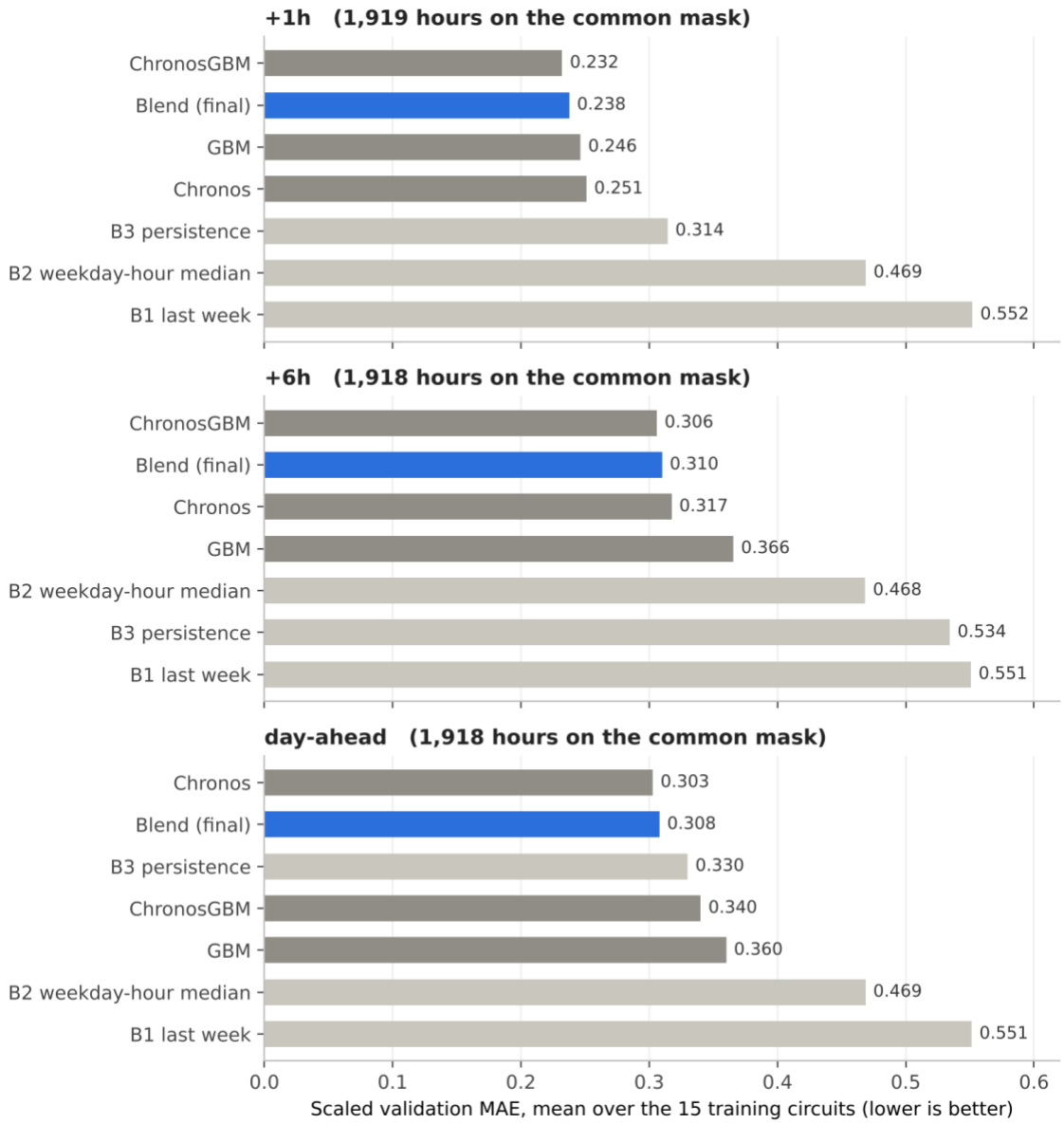
Change from v1.2

Model	+1h v1.2 → v1.3	+6h v1.2 → v1.3	day-ahead v1.2 → v1.3
GBM	0.2816 → 0.2824	0.4318 → 0.3975	0.4294 → 0.3970
Chronos-Bolt	0.2876 → 0.2832	0.3525 → 0.3554	0.3661 → 0.3477
ChronosGBM	0.2825 → 0.2665	0.3885 → 0.3461	0.4032 → 0.3762
Blend	0.2764 → 0.2712	0.3525 → 0.3488	0.3661 → 0.3513

The larger grid helps most where v1.2 was weakest: ChronosGBM at +6h and day-ahead, the GBM at +6h and day-ahead, and Chronos at day-ahead. Where a v1.3 value is slightly higher than in v1.2 (GBM +1h, Chronos +6h), the v1.2 configuration was in the tie and D10 chose a configuration with lower RMSE.

Final comparison on validation

All seven models scored on the same hours (common mask). The final blend is second at +1h and +6h behind ChronosGBM and second at day-ahead behind Chronos, each time by less than 0.006, and clearly ahead of every baseline.



Blue = model chosen by D10. Dark grey = other ML models. Light grey = baselines (tracked, not candidates).

*results/*_validation_predictions_v1.3.parquet · validation 22–28 Sep 2025 · common mask: 1,919 hours at +1h, 1,918 at +6h and day-ahead*

Horizon	Model	Own forecasts, 15 training circuits	Scaled RMSE (own)	Common mask, 15	Common mask, core-8	Common mask, all 16
+1h	ChronosGBM	0.2665	0.6197	0.232	0.244	0.228
+1h	Blend (final)	0.2712	0.6078	0.238	0.246	0.229

Horizon	Model	Own forecasts, 15 training circuits	Scaled RMSE (own)	Common mask, 15	Common mask, core-8	Common mask, all 16
+1h	GBM	0.2824	0.6341	0.246	0.256	0.239
+1h	Chronos	0.2832	0.6142	0.251	0.256	0.243
+1h	B3 persistence	0.3586	0.7979	0.314	0.292	0.304
+1h	B2 weekday-hour median	0.5222	0.9567	0.469	0.429	0.463
+1h	B1 last week	0.5521	0.8864	0.552	0.580	0.549
+6h	ChronosGBM	0.3461	0.7185	0.306	0.314	0.301
+6h	Blend (final)	0.3488	0.7059	0.310	0.315	0.300
+6h	Chronos	0.3554	0.7113	0.318	0.331	0.304
+6h	GBM	0.3975	0.7509	0.366	0.325	0.362
+6h	B2 weekday-hour median	0.5211	0.9535	0.468	0.431	0.464
+6h	B1 last week	0.5507	0.8838	0.551	0.580	0.550
+6h	B3 persistence	0.5723	1.1565	0.534	0.461	0.512
day-ahead	Chronos	0.3477	0.7059	0.303	0.310	0.302
day-ahead	Blend (final)	0.3513	0.7027	0.308	0.305	0.302
day-ahead	B3 persistence	0.3708	0.7358	0.330	0.368	0.316
day-ahead	ChronosGBM	0.3762	0.7278	0.340	0.326	0.335
day-ahead	GBM	0.3970	0.7497	0.360	0.317	0.355
day-ahead	B2 weekday-hour median	0.5217	0.9549	0.469	0.429	0.464
day-ahead	B1 last week	0.5514	0.8851	0.551	0.580	0.550

The common mask drops about 20 % of the valid hours, mainly because B1 needs the same hour one week earlier. The ranking of the ML models is the same as on their own forecasts.

Results per circuit

Scaled validation MAE of the final blend on the common mask. The "best" column names another model (incl. baselines) only where it beats the blend on that circuit ("same" = the blend is best).

Circuit	Hours (+1h)	Training mean (Wh/h)	+1h: Blend	+1h: best	+6h: Blend	+6h: best	day-ahead: Blend	day-ahead: best
NE HVAC	140	2.6	0.182	ChronosGBM 0.176	0.182	GBM 0.177	0.187	GBM 0.186
NE Plugs & Sockets	140	191.5	0.029	Chronos 0.021	0.022	Chronos 0.021	0.024	Chronos 0.023
NW Emerg. Lighting	140	10	0.051	Chronos 0.047	0.055	Chronos 0.051	0.048	Chronos 0.048
NW HVAC	140	124.2	0.017	B3 0.011	0.045	B3 0.022	0.059	B3 0.008
NW Hand Dryer	140	0.9	0.876	ChronosGBM 0.832	0.976	ChronosGBM 0.943	1.030	ChronosGBM 0.949
NW Lighting	140	46.9	0.465	ChronosGBM 0.446	0.736	GBM 0.705	0.652	same
NW Plugs & Sockets	140	563.3	0.335	Chronos 0.312	0.491	same	0.427	Chronos 0.413

Circuit	Hours (+1h)	Training mean (Wh/h)	+1h: Blend	+1h: best	+6h: Blend	+6h: best	day-ahead: Blend	day-ahead: best
NW RACK	140	185.7	0.013	ChronosGBM 0.012	0.016	same	0.016	same
SE Emerg. Lighting	115	2.3	0.220	Chronos 0.217	0.213	same	0.229	same
SE HVAC	115	46.9	0.025	ChronosGBM 0.024	0.023	Chronos 0.023	0.021	Chronos 0.021
SE Lighting	114	129.3	0.412	same	0.760	Chronos 0.710	0.819	B3 0.381
SE Plugs & Sockets (cold-start)	13	255.2	0.097	same	0.143	Chronos 0.108	0.208	B3 0.111
SW Emerg. Lighting	118	11.9	0.054	ChronosGBM 0.051	0.060	Chronos 0.059	0.060	Chronos 0.059
SW Hand Dryer	111	0.9	0.806	ChronosGBM 0.742	0.885	GBM 0.880	0.879	Chronos 0.877
SW Lighting	95	43.1	0.040	ChronosGBM 0.031	0.099	ChronosGBM 0.071	0.092	B3 0.060
SW Plugs & Sockets	118	265.1	0.041	Chronos 0.041	0.087	GBM 0.066	0.077	ChronosGBM 0.062

- **Level changes:** NW HVAC (standby in the validation week), SE Lighting and SW Lighting are still forecast best by persistence (B3) at day-ahead. On SE Lighting the gap is large (0.819 vs 0.381).
- **Hand dryers:** about 0.9 Wh per hour with short spikes. Every model has a scaled error near 1 there, so they weigh heavily in the mean.
- **Cold-start SE Plugs & Sockets:** only 13 hours remain on the common mask (B1 needs the hour one week earlier, which is mostly flagged). Report it separately and on its own forecasts.
- **Fewer hours for SE/SW:** 95–118 common-mask hours against 140 for NE/NW. Their scores rest on fewer hours.

Interpretation, limitations and next steps

With 591 combinations per horizon, combining the pretrained model with the trained one now pays off: ChronosGBM is the best single model at +1h and +6h, and every final model beats the best baseline clearly. Day-ahead remains the hardest horizon. Chronos with a short context leads there, and the GBM-based models stay behind persistence.

Why small trees and short contexts help: when a circuit changes level in the validation week (NW HVAC, SE Lighting), models that learn many training-period patterns pull back towards the old level. Small, regularised trees and a 72 h Chronos context follow the new level faster.

Effect of D10: inside the tie margin D10 prefers the lower RMSE. A mix of several models smooths large errors and almost always has the lower RMSE, so the rule chose the blend at every horizon, at a cost in MAE of 0.0047 (+1h), 0.0027 (+6h) and 0.0036 (day-ahead) against the best single model. This follows the rule agreed before the runs and is not changed after seeing the results.

Limitations

- One validation week only. Differences below about 0.005 are not reliable, and with 591 combinations the best validation score is optimistic. The test week will show how much of the gain holds.

- Several chosen configurations lie at the edge of the grid: 7 leaves (GBM +6h and day-ahead, ChronosGBM +6h), 63 leaves and min leaf 100 (ChronosGBM +1h and day-ahead), context 24 h (Chronos +6h). Smaller or larger values might do better. The grid is not extended after seeing the results.
- The selection mean is dominated by the two hand dryers, whose scaled error is near 1 for every model.
- The linear model (Sprint 3) was not re-tuned on v1.2/v1.3 and is not part of this comparison. Weather data are not yet used (D12).

Next steps (point 8)

1. Upload the v1.3 code, frozen configs and `model_choice_v1.3.json` to the repository and share the model choice with the team before any test forecast is made.
2. Optional D12: add weather inputs to the GBM parts of the chosen blend and check them on validation only.
3. Update `run_task8_test.py` and `run_chronos_test.py` to v1.3: Chronos size and context per horizon, ChronosGBM with the test Chronos forecasts added to the cache, three-model blend with the frozen weights.
4. Run all frozen models once on the test week and report all-16, core-8, per circuit and cold-start separately, with the baselines alongside. Do not change any setting afterwards.

Source of all figures: v1.3 runs of 29 Oct 2026 (Python 3.13.15, scikit-learn 1.9.1, chronos 2.3.2, torch 2.15.0), frozen configs and validation prediction files, recomputed independently. Selection rule D10 agreed 29 Oct 2026 (Emily Wiesner, Alex).